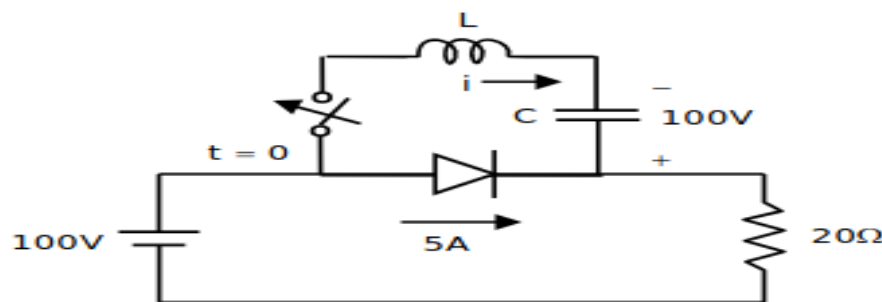


GATE QUESTION ECE 2010 Q55

Question :

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Q55) The L-C circuit of Q54 is used to commutate a thyristor, which is initially carrying a current of 5A as shown in the figure below. The values and initial conditions of L and C are the same as in Q54. The switch is closed at $t = 0$. If the forward drop is negligible, the time taken for the device to turn off is



1. $52\mu\text{s}$
2. $156\mu\text{s}$
3. $312\mu\text{s}$
4. $26\mu\text{s}$

Given from Q55 and Q54:

- Inductance, $L = 1\text{mH} = 1 \times 10^{-3} \text{ H}$ (from Q54)
- Capacitance, $C = 10\mu\text{F} = 10 \times 10^{-6} \text{ F}$ (from Q54)
- Initial Thyristor Current, $I_{\text{initial}} = 5\text{A}$
- Source Voltage, $V_S = 100\text{V}$

Calculation with Given Values: 1. Calculate the resonant angular frequency ω_0 :

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(1 \times 10^{-3} \text{ H}) \times (10 \times 10^{-6} \text{ F})}}$$

$$\omega_0 = \frac{1}{\sqrt{10 \times 10^{-9}}} = \frac{1}{\sqrt{1 \times 10^{-8}}} = \frac{1}{1 \times 10^{-4}} = 10000 \text{ rad/s}$$

2. Calculate the peak commutation current $I_{\text{peak.commutation}}$:

$$I_{\text{peak.commutation}} = V_S \sqrt{\frac{C}{L}} = 100 \text{ V} \sqrt{\frac{10 \times 10^{-6} \text{ F}}{1 \times 10^{-3} \text{ H}}}$$

$$I_{\text{peak_commutation}} = 100\sqrt{0.01} = 100 \times 0.1 = 10 \text{ A}$$

3. Calculate the time t to turn off:

$$t = \frac{1}{\omega_0} \arcsin \left(\frac{I_{\text{initial}}}{I_{\text{peak_commutation}}} \right)$$

$$t = \frac{1}{10000 \text{ rad/s}} \arcsin \left(\frac{5 \text{ A}}{10 \text{ A}} \right)$$

$$t = \frac{1}{10000} \arcsin(0.5)$$

We know that $\arcsin(0.5) = \frac{\pi}{6}$ radians.

$$t = \frac{1}{10000} \times \frac{\pi}{6}$$

$$t = \frac{\pi}{60000} \text{ s}$$

To express this in microseconds:

$$t \approx \frac{3.14159265}{60000} \text{ s} \approx 5.23598 \times 10^{-5} \text{ s}$$

$$t \approx 52.36 \mu\text{s}$$

Final Answer:

The calculated turn-off time is approximately $52.36 \mu\text{s}$, which corresponds to option (A) $52 \mu\text{s}$.